

FOLLOW UP ACTIVITIES

1. Draw a diagram of vernier caliper.
2. Find out diameter of a cylinder by using vernier caliper.

Signature of
Guide Teacher

Signature of
Student Teacher

Review Question

1. Define Lesson plan. Write the need and importance of Lesson Plan.
2. Explain the steps followed to prepare a lesson plan.
3. Explain in detail the Memory Level of Teaching. Write its Merits and Demerits.
4. Explain in detail the Understanding Level of Teaching. Write its Merits and Demerits.
5. Explain in detail the Reflective Level of Teaching. Write its Merits and Demerits.
6. Define a Unit Plan. What are the steps to be followed to perform a unit plan. Write its advantages.
7. Write the difference between Unit plan and Lesson plan.
8. Explain in detail the structure of a four-fold lesson plan.
9. Write a model lesson plan for IX Std. Physical Science.
10. Write a model lesson plan for VIII Std. Physical Science.

UNIT - IV

METHODS OF TEACHING PHYSICAL SCIENCE

Methods of teaching

A method is "a scientific way of presenting the subject keeping in mind the psychological and physical requirements of children."

A Physical Teacher has a variety of methods and techniques available for the use in teaching Physical Science. The selection of suitable method depends upon the objectives of the lesson, nature of the content and needs of the learner. Some methods are more appropriate for teaching students as a group where as some methods are specially designed for individual instruction. A method best for one teacher and applicable for a class, may totally be a failure for another teacher to teach the same other class. So it should be left for the teacher from his wide information, which of the methods to use and when. Here we are going to discuss in detail the various methods available to teach Physical science.

4.1 Teacher-centred methods

In these methods the teacher role is to give knowledge and full control of the classroom.

4.1.1. Lecture Method

It is the oldest teaching method. It is the most dominating method and is liked by the majority of Teachers. In this method teacher present the subject matter and the

students listens. Students are not participating in this method. This method is teacher centred and information centred and in this method teacher works as a sole resource in classroom induction. Due to lack of participation, students get bored and some of them may go to sleep. Here teacher is presenting ready made knowledge and due this spoon feeding. The students' loose interest and their power of reasoning and observation get no stimulus.

In this method teacher goes ahead with the subject matter at his own speed. The teacher may use blackboard at times and may also dictate notes. This teacher centred method in its extreme form does not expect any question or response from the students. It helps only intelligent students who can understand and remember the facts told by the teacher, which is just a blind exercise.

In this method, teacher prepares his lesson plan at home and delivers it out in the class. This method does not cater for releasing the teaching Physical science and is not in accordance with the principles of teaching.

Guidelines for preparing a Lecture

To make lecture method effective, the teacher has to follow certain guidelines while preparing the lecture.

- Lecture should be carefully and systematically planned as lesson plan
- Teaching objectives should be kept in mind while preparing the lecture.
- The lecture should exhibit the thorough knowledge of the subject matter, its organisation, development, interpretation and application.

- Lecture should be related with day to day life experience of the students.
- Student participation should be encouraged by asking questions, clarifying doubts and reviewing the key points at frequent intervals.

Merits of Lecture method

1. It is possible of handle a large number of students at a time and no laboratory, equipment, aids are required.
2. Using this method the knowledge can be imparted to the students quickly and the prescribed syllabus can be covered in a short time.
3. Teachers feel secure and satisfied.
4. Using this method it is quite easy to impart factual information, historical anecdotes and life history of eminent personalities.
5. Teacher can develop his own style of teaching.
6. In this method teacher can easily the logical sequence of the subject by planning his lecture.
7. This method gives the students as well as the teacher a sense of satisfaction and achievement.
8. Lecture method trains students to listen long duration.
9. It delivers a lot of information in a short amount of time.

Demerits of Lecture method

1. In this method students participation is negligible and students become passive recipients of information.

2. We are not sure if the students are concentrating and understanding the subject matter.
3. This method leads weak students develop a hatred for learning.
4. If does not take into account the previous knowledge of the students.
5. If does not cater to the individual needs and differences of the students.
6. It is the lowest retention value of all teaching techniques.
7. It is an undemocratic and authorisation method.
8. In ability to understand one essential point may make the rest of the lecture unintelligible.
9. This method does not help in developing problem solving skills.
10. This method can be used only in higher secondary level or college level.

4.1.2. Demonstration Method

Demonstration method is a method of teaching such subjects that need live guidance on the part of the teacher. Demonstration involves showing by reason, explaining or making clear by use of examples or experiments. This strategy is applied mainly in teacher education programmes. Demonstration often occur when students have a hard time connecting theories to actual practice or when students are unable to understand application of theories. The demonstration method of teaching shows learners how to perform the task step by step so that the learner will eventually be able to complete the same task independently.

Demonstration method focus to achieve psychomotor and cognitive objectives. It is given in three successive steps.

1. **Introduction:** In this step objectives of the lesson are stated. The teacher demonstrates the activity before the student that is to be developed.
2. **Development:** Students try to initiate the demonstrated activity. If there is any query the teacher tries to satisfy them by further demonstration and illustrations.
3. **Integration:** At this step, the teacher integrate all the activities and then these activities are rehearsed revised and evaluated.

Rules for using the demonstration method

1. Give a perfect demonstration.
2. Give a step by step explanation - Give reason, examples and comparisons to make the explanation clear.
3. Continue until the student has imitated each step.
4. Provide student practice, with assistance as necessary.
5. Complete the exercise with an evaluation.

Advantages of Demonstration method

1. It helps in involving various sense to make learning permanent.
2. Teacher gets cooperation of students in teaching learning process.

3. It develops interest in students and motivates them for their active participation.
4. It helps in achieving psychomotor objectives.
5. Even complex skill becomes easy to understand.

Disadvantages

1. It can be used only for skill subjects.
2. Only the attention of the learners are invited towards the activity demonstrated. They are not free to discuss about it.
3. Students are not able to create handmade models for demonstration.
4. It is a time consuming method and the teachers not able to complete the syllabus in time.
5. The Demonstration method is restricted to only certain kinds of teaching content.

4.1.3. Team Teaching

T	-	Together
E	-	Everyone
A	-	Achieves
M	-	More

Definition :

All arrangements that include two or more faculty in some level of collaboration in the planning and delivery of a course materials to a class. Team teaching has now found in different departments ranging from school to college level.

It can help create a dynamic and interactive learning environment. In ordinary class teaching only one faculty covers entire course material which is interested or uninterested to the students.

But in team teaching course material is divided according to their specialisation and interest. Students have the opportunity to hear multiple perspectives. Different faculties create interest, keep attention and prevent boredom.

Types of Team Teaching

There are different approached in team teaching. Some of the more common are :

- Interactive team teaching - two faculty members present in front of the class simultaneously.
- Rotational team teaching- faculty members take classes in rotation. This rotational format has a number of variations depending on the subject matter and the number of faculty involved.
- Participant - Observer Team teaching - all participating faculty are present for all the classes but only one is "teaching" at a time. Other teachers could play as participating observer.

Benefits for faculty from Team teaching

1. Improve their own teaching skills
2. Avoid the lonely repetitive fragmented experience of solo teaching.
3. Have opportunities for creative assignments.

4. Develop clearer perspective on the difference between disciplines.
5. Build bridges of understanding across disciplines.

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Disadvantages

1. Some teachers are rigid personality type or may carry to a single method of teaching.
2. Some simply dislike the other teachers on the team.
3. Allocation of course material to different faculties may sometimes inconvenient.
4. Since two or more faculties present in the class, expenses are more.
5. Teams don't have adequate common planning time.

4.2 Learner-centred methods

In these methods student gather knowledge through discussion, critical thinking, problem solving etc. Teacher's role is to coach and facilitate.

4.2.1 Self-Learning

Self-learning is defined as: "a process by which individuals take the initiative, with or without the assistance of others, in diagnosing their learning needs, formulating learning goals, identifying human and material resources for learning, and evaluating learning outcomes". Self-learning skills involve to manage learning tasks, without the help of others. These skills are necessary for effective lifelong learning and students are expected to develop this skill. It is the modern way of Learning, and it has proven to be more effective and convenient. Now-a-days you can learn how to do anything with a internet.

Problem Solving Method and Peer tutoring are good examples that capture the essence of self-directed learning. A learning contract is commonly used as a tool to assist students in planning for their learning goals and learning actions. Self-learning gives you the ability to identify problems and quickly look for effective solutions on your own.

Importance of self-learning

Self-learning has proved it's worth a hundred times over as it keeps the individual happy, motivated and engaged. It works as a handy tool that leads to high performance and efficiency levels. It is useful for individuals who do not have enough time to enroll themselves in formal courses because of some reason or other.

When a learning process is not forced and the student wants to take the first step at self-learning voluntarily then it is a move in the right direction. Self-learning has to happen from within because you cannot force someone to do it effectively. It is actually motivated by a desire to make yourself richer in knowledge so that you can better your circumstances.

Self-learning is considered enriching and worthwhile because the learner can set a pace that is convenient and affordable. With the advent of the internet, it has become the process of self-learning has become very easy and flexible. It is especially beneficial for professionals who want to brush up their skills to refresh and hone their knowledge.

Self-learning helps a person in understanding the basic concept of learning and it says that everyone has to learn by himself at the end of the day. As you go deeper you start exploring new horizons that might have been a taboo in traditional form of learning and this can open further doors. Actively pursuing your goals single-handedly gives the learner the necessary confidence to deal with the eventualities of life by any means.

Self-learning materials

Any learning resource that can be used by a learner without the presence of a teacher. There are 10 of the best free learning tools currently available for self-learners. They are.

1. Khan Academy
2. Duolingo
3. Solo Learn
4. Coursera
5. Book boon
6. The University of Oxford
7. edX
8. iTunes U
9. Codecademy
10. Udemy

These resources and learning tools can help you boost your personal development and abilities.

Advantages of self-learning

1. It is not time-bound and is dependent upon the person who wants to learn for whatever number of hours he feels like. It gives him the opportunity to set his own pace and remove frustration and boredom from the equation
2. Learning is without any restrictions
3. Self-learning is not subjected to location-constraint as an individual can make use of the learning process from anywhere he pleases. He does not need to travel at a particular place at a specific time and this works in his favor
4. It boosts the self-esteem of a person because he knows that he has learned everything on his own. It also helps him to keep digging until he knows all there is to know about the topic.
5. Self-learning helps to identify issues and find solutions quickly because you do not want to waste your valuable time sorting out issues. The ability to learn, identify and change becomes more spontaneous
6. The importance of self-learning is that it takes the crutches away. You are no longer dependent on others and this proves beneficial in the long run. Identifying and tackling problems works as a booster and gives you immunity against issues and problems that a person faces in life.
7. Self-learning helps you to find the ground under your feet. It prepares you for the future where you have to work alone to reach your goal. The resilience and the immunity that you gain here are the key tools that will take you further in life.

Disadvantages of self-learning

1. You do not have readymade material at your disposal and whatever you need will have to be acquired through additional effort on your part
2. You will have to verify your study materials whereas in traditional learning method these were already verified beforehand
3. Working with others help in brainstorming which is not possible during self-learning
4. Teamwork encourages open communication that is not seen during the process of self-learning
5. There is no face-to-face interaction during self-learning
6. The lack of transformational power is a serious limitation of self-learning

4.2.2 Programmed Instruction

Programmed Instruction is also known as programmed learning as learning is focused in it. It provides opportunity to the learner to study and learn individually.

Learning opportunity is provided through a frame or series of frame having i) learning material ii) in built mechanism of evaluation and iii) related answer of the evaluation part to provide feedback. It is considered the antecedent of modern computer-aided learning.

Principles of programmed instruction

- Principle of small steps
- Principle of Active responding
- Principle of immediate feedback
- Principle of Self-Pacing
- Principle of Student Testing

Programmed instruction is a method of individualized instruction where each individual learns by himself at his own rate. Programme is a sequence of small steps of instructional material (called frames) most of which requires a response to be made by completing a blank space in a sentence. A system of cueing is applied and each response is verified by the provision of immediate knowledge of results. The teacher is not physically present. The programmer while developing programmed materials has to follow the laws of behavior and validate his strategy in terms of student learning. It is based - on skinner's operant conditioning.

Characteristics of Programmed instruction

- It is an individualized instruction
- Materials are developed keeping in view logical sequence
- Immediate feedback is being provided
- There is inbuilt mechanism of interaction between the learning materials and learners.
- Learners learn the material with their own pace
- Continuous evaluation is integral part of the learning material.

Development of Programmed Learning Material

- i. Preparatory phase
 - ii. Development or writing phase
 - iii. Evaluation phase / Tryout and Revision
- i. Preparatory phase**
- Selection of a unit or topic
 - Preparation of content outline

- Defining objectives in Behavioural terms
- Construction of a Test of Entering behavior
- Construction of Test of Terminal behavior

ii. Development or writing phase

- Present the material in Frames
- Provide for Active Student Response
- Constructing test after each frame
- Provide for correction of student Response
- use of prompts to Guide student Response
- Provide careful sequencing of Frames
- Preparing answer Key

iii. Evaluation phase

- To evaluate to improve through pre-tryout and tryout.
- To Revise
- To evaluate the impact of the learning material

Advantages of programmed Learning

1. There is no fixed time for learning.
2. Students may learn at their own pace.
3. Programmed learning is an strategy for self-learning easily without any tutor.
4. It gives drill, practice, self-evaluation and motivation
5. It is a new method of presenting context to students through consequent sequences in a controlled manner.
6. It enhances students critical thinking ability and power

7. In short period of time students can learn huge content.
8. It helps for individual learning and avoids social and emotional problems.
9. It immediately provides result of progress
10. It is very helpful for distance learning.

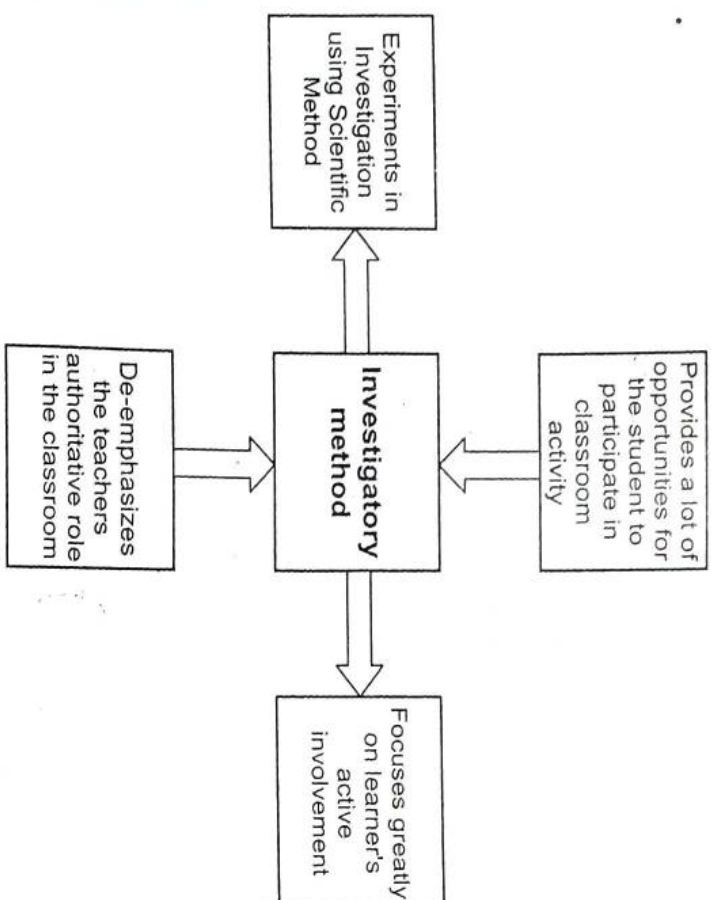
Disadvantages of programmed Learning

1. It limits students in creativity and originality
2. Preparation of material is time consuming
3. It restricts student teacher interaction and relationship
4. Students may get separated from society
5. Doubts cannot be resolved through this instruction immediately as formal education.

4.2.3 Investigatory approach

"The investigative approach is a method of learning and teaching that gives students opportunities to direct their own learning as they explore the science underlying realistically complex situations." ... Students investigate scientific problems that they find meaningful. Any kind of activity that involves getting to the truth or gathering facts is called investigatory approach.

Inquiry-based learning is a learning and teaching approach that emphasizes students' questions, ideas and observations. This form of learning enhances comprehension - rather than memorizing facts and taking notes. Students are encouraged to discuss ideas among their peers.



Investigate approach is a style of pupil learning which gives student strategies and process skills which they can apply in other situations whether it is physical science or in other curriculum areas. Investigative skills are the abilities individuals use to gather data and generate conclusions to reveal important facts.

Role of a teacher in the investigative approach

Teachers are effective in mentoring students in identifying the problem, formulating the hypothesis, writing the review, constructing the research design, gathering of data, data analysis, presentation and interpretation.

- The teacher provide information to the students about the source of material and how to use it.

- Teacher support in many forms such as a coach, counselor etc.
- Teacher monitor the progress of students and encourages in problem solving.
- Teachers help students to connect their previous knowledge to an investigation links.

Techniques of Investigator approach

1. Laboratory Technique :

Students actively manipulates and study a given problem and finds a solution scientifically.

2. Problem Solving Technique :

Learner work activity in the solution of a difficulty or an undesirable situation.

3. Research Technique :

It is a careful and an organized study designed to serve a specific purpose. Doing a simple experiment.

4. Experimenting Technique :

Student investigate a given situation by being a part or an integral component of it and preparing report.

5. Field - study Technique :

It is a methodical trial or error procedure carried out with the goal of verifying, falsifying establishing the validity of a hypothesis.

Tips for improving Investigative skills

Hence are some ways in which you can improve your investigative skills.

a) Find multiple sources

When investigating, it is important for students to seek information supported by more than one source. This can help you ensure the factuality of evidence. Multiple sources also allow you to defend your findings when preparing a report.

b) Maintain neutrality

To avoid biases in your investigations, it is also essential to maintain a neutral stance. This often involves separating your personal beliefs from your investigations.

c) Practice interviewing

In order to improve your interviewing, consider questioning family members or friends. Conducting this can help you develop confidence when interviewing.

d) Pace your investigations

Prior to analyzing the causes of a situation, it is important to gather all relevant evidence before progressing. It is also essential to analyse the collected data to determine whether there is relationship between events.

4.2.4 Collaborative learning

Collaborative learning is based on the view that knowledge is a social construct. Collaborative activities are most often based on four principles.

- The learner or student is the primary focus of instruction.
- Interaction and "doing" are of primary importance
- Working in groups is an important mode of learning.
- Structured approaches to developing solutions to real-world problems should be incorporated into learning.

Collaborative learning can occur peer-to-peer or in larger groups. Peer learning or peer instruction is a type of collaborative learning that involves students working in pairs or small groups to discuss concepts, or find solutions to problems. This often occurs in a class session after students are introduced to course material through readings or videos before class, and/or through instructor lectures.

Benefits of collaborative learning

Research shows that educational experiences that are active, social, contextual, engaging, and student-owned lead to deeper learning. The benefits of collaborative learning include:

- Development of higher-level thinking, oral communication, self-management, and leadership skills.
- Promotion of student-faculty interaction.
- Increase in student retention, self-esteem, and responsibility.
- Exposure to and an increase in understanding of diverse perspectives.
- Preparation for real life social and employment situations.

What are some examples of collaborative learning or group work activities?**Stump your partner**

- Students take a minute to create a challenging question based on the lecture content up to that point.
- Students pose the question to the person sitting next to them.

- To take this activity a step further, ask students to write down their questions and hand them in. These questions can be used to create tests or exams. They can also be reviewed to gauge student understanding.

Think-pair-share/ Write-pair-share

- The instructor poses a question that demands analysis, evaluation, or synthesis.
- Students take a few minutes to think through an appropriate response.
- Students turn to a partner (or small groups) and share their responses. Take this a step further by asking students to find someone who arrived at an answer different from their own and convince their partner to change their mind.
- Student responses are shared within larger teams or with the entire class during a follow-up discussion.

Catch-up

- Stop at a transition point in your lecture.
- Have students turn to a partner or work in small groups to compare notes and ask clarifying questions.
- After a few minutes, open the floor to a few questions.

Fishbowl debate

- Ask students to sit in groups of three.
- Assign roles. For example, the person on left takes one position on a topic for debate, the person on right takes the opposite position, and the person in the middle takes notes and decides which side is the most convincing and provides an argument for his or her choice.

Case study

- Create four to five case studies of similar difficulty.
- Have students work in groups of four or five to work through and analyze their case study.
- Provide 10-15 minutes (or adequate time to work through the cases).
- Walk around and address any questions.
- Call on groups randomly and ask that students share their analysis. Continue until each case study has been addressed.

4.2.5 Experimental learning

Carl Rogers distinguished two types of learning cognitive and experimental. The former corresponds to academic knowledge and the latter refers to applied knowledge. Experimental learning is the process of learning through experience and is more specially defined as learning through "reflection on doing".

The general concept of learning through experience is ancient. Aristotle wrote, "for the things we have to learn before we can do them, we learn by doing them". In 1970 David A. Kolp helped to develop the modern theory of experimental learning, drawing heavily on the work of John Dewey, Kurt Lewin and Jean Piaget.

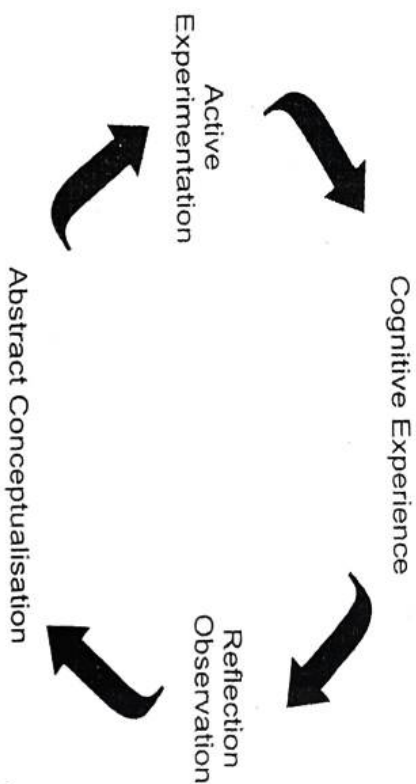
David Kolb's Experimental Learning

Experimental learning can exist without a teacher and solely to the meaning - making process of the individual's direct experience. However a genuine learning require certain elements. According to Kolb, knowledge is continuously gained through both personal and

environmental experiences. Kolb states that in order to gain genuine knowledge from an experience, the learner must have four abilities.

- The learner must be willing to be actively involved in the experience.
- The learner must be able to reflect on the experience.
- The learner must possess and use analytical skills to conceptualize the experience and
- The learner must possess decision making and problem solving skills in order to use the new ideas gained from the experience.

Four steps of experimental learning



1. **Concrete Experience:** A new experience of situation is encountered or a reinterpretation of existing experience.
2. **Reflective Observation:** Reviewing or reflecting on the experience observation and reflection on that experience.

3. **Abstract conceptualization:** Reflection gives rise to new idea or a modification of an existing abstract concept and leads to generalizations.

4. **Active Experimentation:** The learner used to test hypothesis in future situations, resulting in new experiences.

Effective learning only occurs when a learner is able to execute all four stages of the model.

4.2.6 Computer Assisted Instruction (CAI)

Computer Assisted Instruction is an interactive instructional technique whereby a computer is used to present the instructional material and monitor the learning that takes place. It may be used in offline / online. It is a self learning technique whereby the computer is used to provide learning resources, record keeping, progress tracking and assessment of learner performance.

CAI uses a combination of text, graphics, sound and video in enhancing the learning process. The computer has many purpose in the classroom, and it can be utilized to help a student in all areas of the curriculum. CAI helps to improve instruction. CAI programs are tutorials, drill and practice, simulation and problem solving approaches to present topics and they test the students' understanding. They will also give notes on incorrect responses. Generally it involves three types of techniques namely hardware, software and courseware

Types of Computer Assisted Instruction

1. **Drill and practice :** Drill and practice provide opportunities in skills to become mastery in the content.

2. **Tutorial** : Tutorial activity includes both the presentation of information and its extension into different forms of work, including drill and practice games and simulation.
3. **Games** : Content in the forms of Games are presented. Games software help to achieve highest score.
4. **Simulation** : Simulation software can provide an approximation of reality.
5. **Discovery** : Discovery approach provides a large database of information specific to a course or content area and challenges the learners to analyze, compare, infer and evaluation based on their exploration of the data.
6. **Problem Solving** : This approach helps students develop specific problem solving skills and strategies.

Advantages of CAI

- CAI provide one-to-one interaction
- It is a great motivator
- Instantaneous response / immediate feedback to the answer elicited.
- Self-pacing - Allow students to proceed at their own pace.
- Helps teacher can devote more time to individual students
- Privacy helps the shy and slow learner to learn
- Learn more and more rapidly
- Multimedia helps to understand difficult concepts through multi sensory approach

- Self directed learning - Students can decide when, there and what to learn.
- CAI can be arranged for nearly 4000 students simultaneously.

Disadvantages of CAI

- May feel overwhelmed by the information and resources available.
- Over use of multimedia may divert the attention from the content.
- Learning becomes too mechanical
- Non availability of good CAI packages
- Lack of infrastructure
- CAI can become boring if a student is alone at a terminal for too long time

4.2.7 The Keller Plan

The Keller plan, also called the Personalized System of Instruction (PSI) was developed by Fred. S. Keller and his associates in the middle of 1960 as an innovative method of instruction. This individualized learning method was oriented on improvement of high school learning. It is highly personalised learning method in which students progress at their own pace. It is based on Skinner's theories of operant conditioning.

- Go-at-your-own-pace. A student can move through the course content at his own pace, independently of other students.
- Detailed definitions of learning objectives.

- Unit perfection requirement. In order to advance to the next unit, a student-needs to demonstrate mastery of the preceding unit.
- Emphasis on written materials. The emphasis is on learning from written materials. Lectures and demonstrations will be provided only when you have interest in it. No examination will be based upon them; and you need not attend them if you do not wish.
- The use of proctor. Proctors entrance the social aspect of education process, enable repeated testing with immediate scoring and tutoring. They help with individual problems.

Course module

Physically the teaching Lab has tables with seating for about 20. The lab is open to students for three to four hours each day and one or more tutors are available in the lab each of those hours. Students come to the lab to write their quiz and to seek help with course material from the Tutors or from other students. In the Lab students receive quiz sheets and work individually to answer them. Students may consult their texts, notes and calculators freely during the quizzes, monitoring of the students is not a concern. The quizzes consist of ten multiple choice questions and students must answer eight correctly. Otherwise they must return to the lab for the next attempt. If necessary students take repeated attempts to demonstrate their mastery. Multiple forms of quizzes are prepared for repeated candidates. Result of the quizzes are posted in the lab on the next day.

The modules can consist of reading assignments, films, audio tape, film strips, programmed instruction, conducting an experiment, conducting an interview etc.

Keller divided the process for creating PSI into four steps.

- Determine the material for the course
- Divide the material into self-contained module (segments)
- Create methods of evaluating the material in a given module.
- Allow learners to move from module to module at their own place.

Advantages of Keller Plan

- PSI has significantly positive effects on learning.
- In Keller plan there is no punishments, and the instructors simply award pass or fail grades. It yielded better results than more traditional lecture-based learning.

Disadvantage of Keller Plan

- Limited instructional methods
- High dropout rates
- Decreased human interaction.

4.2.8 Project Method

Project is a purposeful activity related to life and it should be carried out in a natural environment in a school or outside the school. Project method is of American origin and is an outcome of Dewey's philosophy of pragmatism. This method is developed and applied practically by Dr. Kilpatrick in this method student's learn subject matter in natural setting. Here the content knowledge and skills found in mathematical text book are carried out with the actual

happenings in life. Projects can be done individually or in a group or all the students together.

Basic principles of project method

- Learning by doing.
- Learning by living.
- Learning through association, co-operation and activity.
- Learning by experience.
- Learning by sociability

Some projects for mathematics

- Running a co-operative bank in the school.
- Running a stationary store in the school.
- Running a school hostel.
- Lying road in the school.
- Maintaining kitchen garden in the school.

Steps in a project

For completing a project, we have six stages

1. Providing a situation.
2. Choosing a project.
3. Planning of the project.
4. Executing the project.
5. Evaluating the project.
6. Recording all activities of the project.

1) Providing a situation:

A project should arise out of a need felt by students and content of the subject. It should never be forced to them. For this teacher will explain various projects according to need, interest of the students, and their skills to do the project.

2) Choosing a project:

Project will be selected according to the wish of the students. The teacher may give guidance in selection of a better project if the student's make an unwise choice.

3) Planning the project:

The students he encouraged by the teacher to plan out the details of the project. While planning, the following points to be considered. i) the nature and scope of the project. ii) the degree of complexity of project. iii) time allotted to finish the project and iv) availability of material resources. Discussion may be held among students before the final decision is agreed upon.

4) Executing the project:

The teacher helps the student's in assigning work to different student's accordance with their interest, aptitude and capabilities. Each member of the group should be actively involved in the execution of the project. The teacher supervise and guide the students in the execution of the project.

5) Evaluation

The students along with the teacher should review the progress of the project at frequent intervals. The evaluation of the project has to be done in the light of plans, difficulties in the execution and achieved results. Let the students have self criticism and look through their own failings and findings.

6) Record

A complete record of the project be kept by the students. The record should include everything from the beginning to end about the project. It should include the proposed plan, duties allotted to different students and how

far were they carried out by them, details of places visited, guidance we received, books consulted and all other details.

Example 1: Laying out a vegetable garden in the school.

Providing a situation: After knowing the interest, abilities of the students and content. Teacher explain projects like laying road in the school, Lying vegetable garden, Running a stationery store.

Choosing a project: According to wish of the students the project selected is laying out a vegetable garden in the school.

Planning the project: To lay a vegetable garden in the school, the following points are discussed.

- i. Area of the garden and place selected for garden.
- ii. What are the vegetables to be grown ?
- iii. How much of seeds, fertilizers and pesticides we have to buy?
- iv. Type of work to be allotted to students.
- v. Total amount required to form the garden.
- vi. Expected revenue.

Execution: Each member of the group actively involved in the execution of the project. The teacher supervise the work and guide the students then and there.

Evaluation: With the help of project, students learn how to calculate the area of the garden, profit or losses and percentage of profit or loss.

Recording: All activities of the project are recorded here. Area of garden, area of vegetables planted, total expenditure, total revenue, profit or loss, profit or loss percentage guidance for future etc are all recorded here.

Example 2:

Running of a hostel mess in the school
It involve the following steps.

1. The number of hostlers will be recorded.
2. Secretary of the hostel will be selected by the hostlers.
3. The expected expenditure will be calculated.
4. Expenditure on various heads will be allocated.
5. Budget will be prepared.
6. The amount of collections from the students will be noted.
7. A chart of 'balanced diet' for the hostlers will be prepared.
8. The time of breakfast, lunch, tea and dinner will be notified.
9. Weight of each hostler will be checked after regular intervals and the same will be put on notice board.
10. Punctuality in all the activities will be maintained.
11. Evaluation of all the entire program.
12. Entire activities are recorded and will be informed to all concerned.

Merits of project method

1. It is a psychological method and learner's centered.
2. It introduces democracy in education.
3. It promotes social interaction, inculcates spirit of co-operation and exchanges of experiences among the students.

4. It develops self confidence and self-discipline.
5. It makes the learning more interesting and facilitates better understanding of the subject matter as the learning is related to reality.

Demerits

1. The project method is expensive and time consuming and it is not possible to fit into the regular timetable.
2. It does not provide any training in mathematical thinking and reasoning.
3. Systematic and continuous teaching is not possible.
4. The knowledge obtained from various activities are not continuous.
5. All the students are not getting same type of learning.

4.2.9 Activity Based Learning (ABL Method)

To make students to involve in an activity and thereby learning the subject matter with interest is called Activity based learning. Children with the help of materials participate in activities, doing exercises given and fulfill in learning.

In Activity based learning a skill is taken and learn it without any doubt and then only be proceeds to next skill. After knowing all skills of a class, he will be allowed to go next class. Children themselves take the learning cards of a skill and learns step by step.

For the first time in Tamilnadu activity based learning was introduced in 2003 in 13 model schools under 10 regions of Chennai Corporation. Due to success of this method, this method was introduced in all schools of Chennai corporation in 2004-05. From the academic year 2006-07 this method was implemented all over Tamilnadu in elementary schools for the stds 1 to 4. The implementation

of this method in Tamil nadu is due to credit of Thiru M.B Vijaykumar I.A.S.

Approaches of Activity Based Learner

- Following the curriculum completely.
- Each skill is divided into small parts and change it into learning steps.
- In each learning step various activities are followed.
- All learning steps are linked to complete the syllabus.

Activities arranged in each learning step

- Initial activities are formed for skill development.
- Giving motivation for the skill introduced.
- Activities for skill exercises.
- Evaluation activities.
- Diagnostic activities.
- Wholesome activities for the skill.

The four core learning styles of Activity based learning are

1. Visual
2. Auditory
3. Reading and writing
4. Kinesthetic
1. Visual

Visual learners are better able to retain information when it is presented to them, such as arrows, charts, diagrams, symbols and more.

2. Auditory

Auditory learners prefer listening to information that is presented to them vocally. These learners work well in group settings where vocal collaboration is present and may enjoy reading aloud to themselves too.

3. Reading and Writing

Focusing on the written word, reading and writing learners succeed with written information on work sheets, presentations and other text-heavy resources. These learners are note takers and perform strongly when they can reference written text.

4. Kinesthetic

Taking a physically active role, kinesthetic learners are hands-on and thrive when engaging all of their senses during course work. These learners tend to work well in scientific studies due to the hands-on lab component of the course.

Following are few activity based learning methods

1. Real objects :

Real objects work better when trying to understand them rather than something virtual or imaginary. Science lessons can be extremely interesting through this method of teaching. For example let us go for classification of plants. Students can be asked to collect different varieties of plants and the teacher can teach them about the plant during class.

2. Project work :

For projects, students do it themselves or get help from their parents. But whatever it may be, students get to familiarise with the project topic they are working on. Having regular project work not only increases subject knowledge but also increases curiosity to learn more and with clarity.

3. Video classes :

Videos have taken the centre stage in grabbing attention. Any form of video will earn the interest of the students. Even a short video featuring news could be the subject of interest for students who are bored of seeing the blackboard all the time. Arrange video classes to teach the content.

4. Field trip :

Many a times sitting in the same classroom is boring. Changing the environment can do wonders. If you want to teach post-office, railway station, river or a factory take the students to that place and teach them.

5. Story and role plays :

Do you now why language classes are more interesting than the other subjects? It is because of lots of stories and enacting them through role plays. This method could be tried out for other subjects too. It helps in breaking the monotony of the lessons and getting the students animated.

Merits of Activity Based Learning

1. There is no content burden.
2. No examination fear for children.
3. Learns on par with individual difference.
4. Teacher - student relation improves.
5. Learns with peer group.
6. Students who are absent can continue their lessons where they left.
7. Improves with self - Confidence.
8. Backward students improves in learning with the help of gifted students.
9. Teacher can teach speak with individual student.
10. Flexible home assignments.

11. Recognition is given to particular achievements. This create interest to students to learn further.
12. This is a curriculum which develops full personality.

Demerits

1. Teachers work load increases.
2. If the number of students in the class is more than class discipline fails.
3. Learning affected by a teacher who took frequent leave.
4. Evaluation of learning is difficult.
5. Since we are teaching only minimum skills which are required, we are not competent with private schools.

4.2.10 Active Learning Method (ALM)

Active learning is an approach to instruction that involves activity engaged students with the course material through discussions, problem solving, case studies, role plays and other methods. Active learning approaches place a greater degree of responsibility on the learner than passive approaches such as lectures, but instructor guidance is also necessary in the active learning method. Active learning activities may range in length from a couple of minutes to whole class sessions or may take place over multiple class sessions.

Active learning activities help promote higher order thinking skills such as application of knowledge, analysis and synthesis. These activities engage students in deep rather than surface learning and enable students to apply rather than surface learning and enable students to apply and transfer knowledge better.

Principles of Active learning

1. Reflective - Students' reflection on the meeting of what is learned.

2. Negotiated - negotiation of goals and methods of learning between students and teachers.
3. Critical - Student appreciate different ways and means of learning the content.
4. Complex - Students compare learning tasks with complexities existing in real life and making reflective analysis.
5. Engaged - real life tasks are reflected in the activities conducted for learning.

Nature of active learning

Active learning techniques include role-playing, case studies, group projects, think-pair-share, peer teaching debates, teaching and short demonstrations followed by class discussion. In active learning the teacher acts as a guide. The teacher's task in to let the learners lean on their own through the use of different active learning activities. Active learning is the opposite of passive learning. It is learner - centred not teacher - centred. Active participation of each and every student is a necessary aspect in active learning.

Active Learning Techniques

Few active learning techniques are given below :

1. Tyre Think-Pair-Share activity encourage all students to interact with the material. In this activity, the instructor states an open-ended question. Ask students to spend a minute or two thinking about the writing a response. Then ask students to pair with a partner to discuss their responses.
2. Use a one minute paper test in your class as a formative assessment. Give students 1 to 2 minutes to write brief

responses. Address student responses either during next class or online.

3. **Providing Feedback :** Students often don't know why something is wrong, so whenever possible and time permitting, take a few moments to explain why you have marked incorrect on tests and assignment. Give plenty of positive feedback as well as feedback
4. **Cooperative learning :** Students learn effectively when they are working together. Plan activities that require students to work together and learn from one another. In the process they will learn critical thinking skills, communication skills, problem solving skills and more.
5. **Class Discussion :** Another way for students to teach each other is through class discussion. As students take turns discussing the subject, you can assess their knowledge and discover which students grasp the concept and to what extent.

Advantages of active learning methods

1. Research has proven that there is increased content knowledge for participants of the approach.
2. Development of critical thinking and problem-solving are other benefits.
3. Creative thinking, collaborative and interpersonal skills also show great improvement when active learning methods are implemented.
4. Students engage in higher-order thinking tasks such as analysis, synthesis and evaluation.
5. Promote research based learning through investigation
6. Encouraging leadership skills of the students through self development activities.
7. Integration of prior knowledge with new knowledge to incur a rich structure of knowledge among the students.

8. Cultivating a dynamic environment through interdisciplinary learning and generating high-profile activities for a better learning experience.

Activities - based Learning principles for the Digital Classroom

Due to Covid-19 pandemic schools, colleges and universities across the country were forced to accelerate their on live lesson delivery. There are five activity - based learning principles. teachers can bring into their digital delivery.

1. Act as a guide rather than a lecturer

No one wants to state at a talking head for hours. Break class time into short segments. Between each segment provide students with activities such as polling questions or clicker questions that students can answer with their own smart phone. Other ideas for between segment breaks are lecturer provide as on topic quote and students take five minutes to formulate an argument for or against the topic.

2. Maintain open Communication with students

Since lecturers are generally teaching to a webcam, they have no way of picking up the visual clues from students. Using calls and emails, get how well students are learning, suggested activities include, inviting students to design and lead class discussion, asking students to email the instructor a question or idea related to the topic.

3. Encourage student interactions

When students learn only on online, they miss out on learning form their peers. You can implement some strategies to facilitate interaction and get students to exercise their communication skills during and when outside of class session.

4. Be flexible and creative

With the shift to online learning, boredom and tensions can run high. Try to inject some humor into lessons. Be as flexible as you can with your assignment time table.

5. Explore interactive online tools

There are variety of online tools for interactive teaching. Zoom, Panopto, Google hangout, skype and pre-recording teaching resources etc are interactive online tools use the available resources for the students.

4.2.11 Advanced Active Learning Method (AALM)

Advanced Active Learning method is designed to engage a students' critical thinking skills and apply previous and new knowledge to real-life situations. It emphasis on the higher order learning processes.

It is designed for students for their own learning with structured and open-ending guidance from instructors.

Importance of Advanced Active Learning Method

Advanced active learning involves student development of critical thinking and problem solving skills. More advanced engaged learning activities that require additional planning by the instructor are case-based learning, team-based learning or problem-based learning. Students continue to fail in their ability to demonstrate complex thinking skills because their educational experiences have provided little support for development of these skills for maximal performance. Multiple teaching styles exist to encourage students to apply critical thinking skills to open-ended problems utilizing multiple points of view.

Responsibility of the Instructor

The traditional model of students are passive recipients of learning which is to be inadequate to factor a students understanding and enhance a students' interest. As students' become active learners, faculty become active teacher. A teacher's responsibility moves beyond providing

a educative lecture and more towards encouraging students to be more independent after providing a conducive learning environment.

Advanced Active Learning Strategies

Case based learning or scenario-based learning helps students develop a deeper understanding of the material. The instructor is responsible for creating a complex and engaging scenario or case which parallels the concepts of the lesson. Student may be additionally directed using guided questions. Students will use critical thinking skills to analyse the scenario while communicating a collaborating as a group. Using this technique, students are presented with real life scenarios that are multifaced.

Team-based Learning (TBL)

This approach utilizes student-student interaction in small team to develop critical, practical and creative teaching in their courses. Students assume the role of bring inquiries and faculty burnout is decreased with increased students responsibility and student engagement. Key characteristics of TBL include the creation of heterogeneous work groups.

The team based learning module is especially appropriate in larger class sizes by allowing the class to function or multiple small classes.

Problem-based learning

In this approach, a student learning is facilitated by an instructor presented a problem. Students will work through this carefully constructed open-ended problem by engaging and applying new knowledge. The success of PBL is dependent on several factors: Problems should allow for free inquiry, collaboration is encouraged and essential and feed back in the form of reflection on their learning should be provided. The premise of PBL resides in the students to apply previous knowledge to new situations.

For the success of Advanced Active Learning Method

- Allow the student to be the inquirer.

- Think strategically of what your students to accomplish by the end of the class and design your advanced active learning lesson accordingly.
- Be mindful of the time for preparation for an advanced active learning structure. If designed poorly, students may be less engaged.
- Employ an advanced learning model strategically that best fits the structure and purpose of the course.
- Heterogeneous groups should be created with a structure in place to allow each team to function independently.

4.2.12 Concept Map

Concept maps are visual representations of information. It is a diagram that depicts suggested relationships between concepts. Concept map was developed by Joseph D. Novak in 1960. They can take the form of charts, graphic organizers, tables, flowcharts, vendiagrams, timelines or T-charts. Concept maps are especially useful for students, who learn better visually, although they can benefit any type of learner. They are a powerful study strategy because they help you see the big picture. Big pictures makes details more significant and easier to remember.

Concept maps work very well for classes or content that have visual elements or interms when it is important to see and understand relationships between different things. They can also be used to analyse information and compare and contrast.

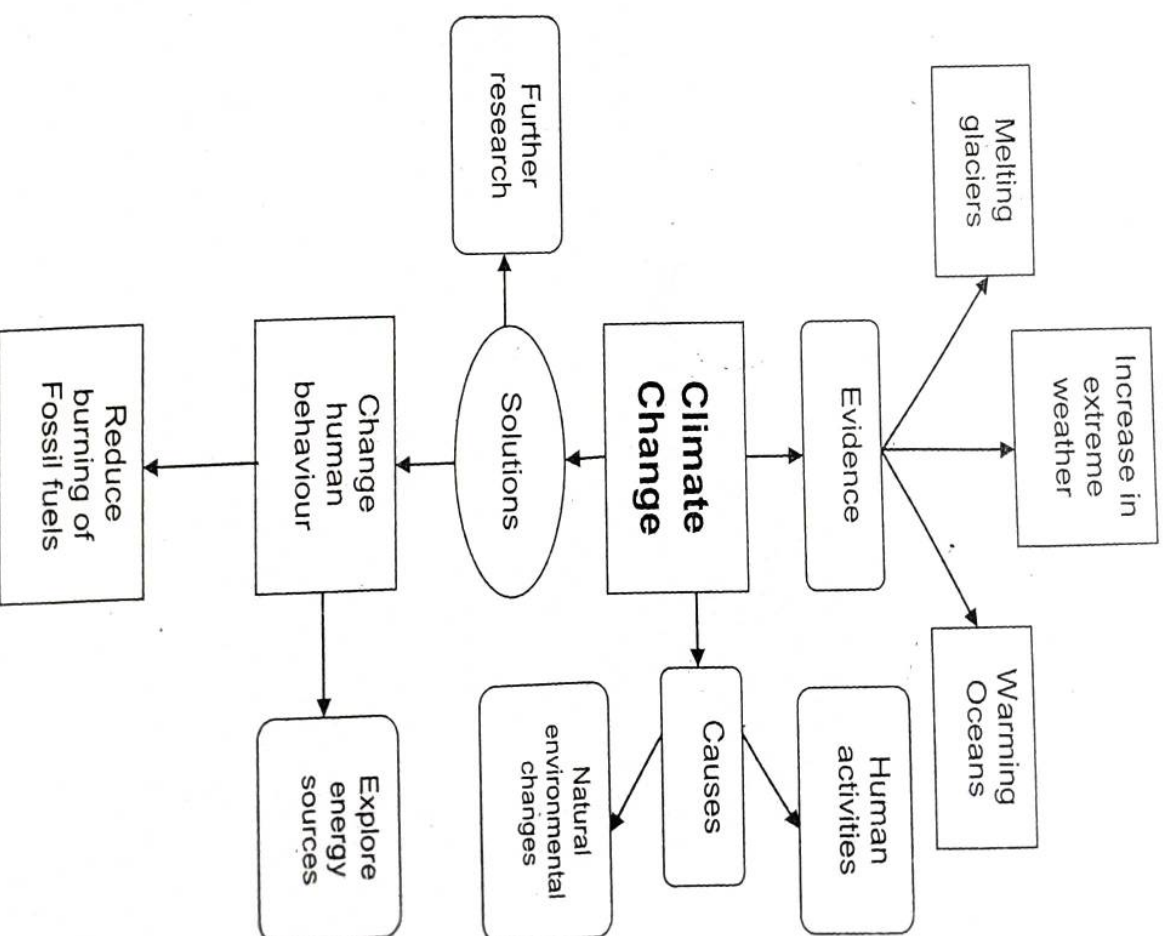
Making Concept Maps

Making concept maps is simple. Follow the steps given below.

- Identify a concept
- From memory, try creating a graphic organizer related to this concept. i.e. Write down subordinate concepts.
- Go through lecture notes, reading and any other resources you have to fill any gaps.

- Focus on how concepts are related to each other.
- Elaborate each part of the Map.

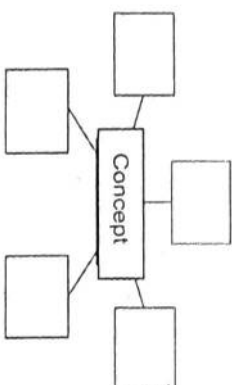
Example of Concept map



Types of Concept Maps

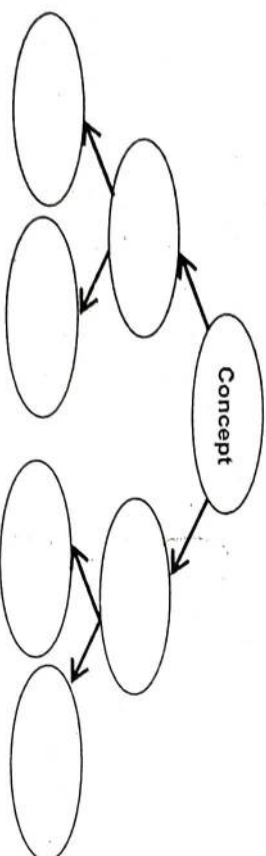
i. Spider Map

All data is organized around a unified theme.



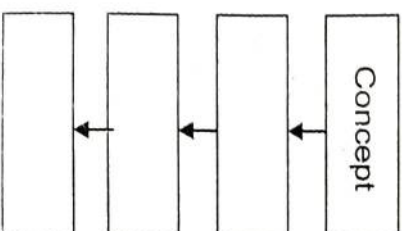
ii. Hierarchical or Chronological

Most general data is located at the top and moves to the most specific.



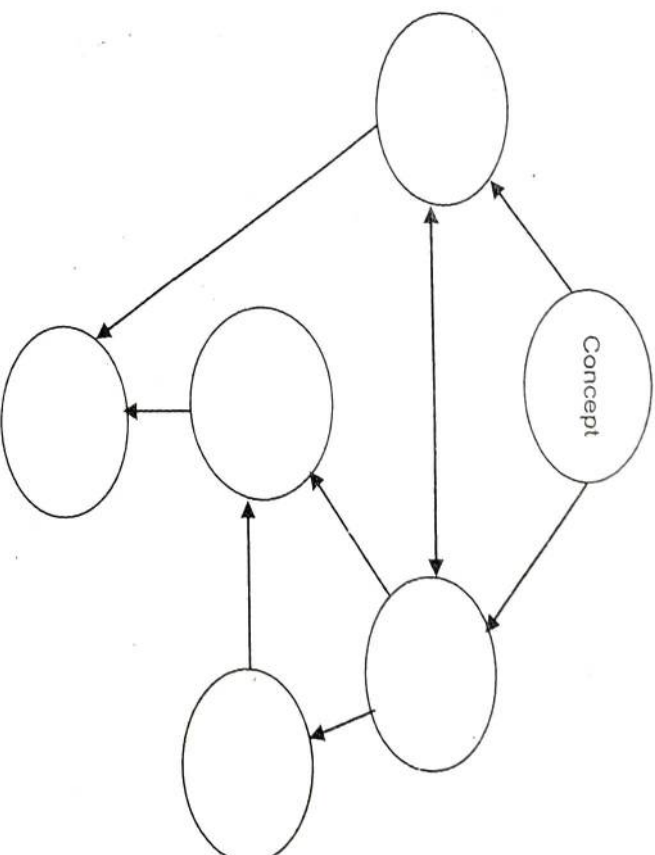
iii. Flow Chart

Information is organized in a logical, ordered manner.



iv. Systems map

Includes all data on the map and shows many relationships between the data.



Uses of Concept maps

- Concept maps are used to stimulate the generation of ideas, and are believed to aid creativity.
- Concept mapping is also sometimes used for brainstorming.
- Concept maps can be used to communicate complex ideas.
- Concept maps are widely used in education and business.
- Note taking and summarizing glean key concepts, their relationships.